ModelIt!

Teacher's Guide

How Do Baby Animals Look Like Their Parents?

The Science of Inherited Traits

Quick Reference

Grade Level	Grade 1
NGSS Standard	1-LS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 1-LS3-1

Make observations to construct an evidence-based account that young plants and animals are like, but not exactly like, their parents.

Three-Dimensional Learning

Science Practice	**Constructing Explanations** — Students use observations of parent and baby animals to explain how traits are inherited.
Disciplinary Core Idea	**LS3.A Inheritance of Traits** — Young animals are very much, but not exactly, like their parents. Plants also are very much, but not exactly, like their parents.
Crosscutting Concept	**Patterns** — Students observe patterns of similarity between parents and their offspring.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Every baby animal in the world looks something like its parents. This is because parents pass special information to their babies that controls what they look like.

What Are Traits?

A trait is any feature of a living thing. Fur color, eye color, ear shape, body size, tail length — these are all traits. Animals inherit most of their traits from their parents. That is why puppies look like dogs and kittens look like cats. The information for these traits is passed from parents to babies before the baby is even born.

Similar But Not the Same

Even though baby animals look like their parents, they are never exact copies. A litter of kittens from the same parents can have different fur colors and patterns. Some might have long tails and others shorter ones. This variation is completely normal and is what makes each animal unique. The differences come from the way parent traits mix together in new combinations.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System

Students identify parent traits as something that already exist (external) and baby traits as something that results from those parent traits (internal).

E — Establish Relationships

Students discover that baby traits are similar to parent traits — when a parent has a certain feature, the baby is likely to have a similar feature.

V — Visualize & Model

Students build a simple model connecting parent traits to baby traits with an arrow.

E — Evaluate Outcomes

Students test their model by predicting what babies might look like based on their parents, then comparing to real photos.

R — Revise & Extend

Students reflect on their own traits and think about which ones they might have inherited from their own families.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at this mama dog and her puppies! Do the puppies look like their mom? How?"*

Watch For

Show a photo of a parent animal with babies. Have students call out similarities they notice.

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn why baby animals look like their parents. We will learn the word TRAIT — that means a feature an animal has."*

Watch For

Point to your own traits: hair color, eye color. Ask students to name their own traits.

Slide 3: Let's Be Scientists!

3 min

Say: *"Why do baby animals look like their parents but not exactly the same? What do you think?"*

Watch For

Show side-by-side photos of parents and babies. Students share what is the same and what is different.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts: Parent Traits are the features the mom and dad have. Baby Traits are the features the baby gets."*

Watch For

Show a parent animal card with labeled traits. Ask: What traits might the baby get?

Slide 5: How Are They Connected?

5 min

Say: *"Parents pass their traits to their babies. When a parent has spots, the baby will probably have spots too! But maybe not in the exact same places."*

Watch For

Students use sentence frame: The baby looks like its parent because it inherited _____.

Slide 6: What Did We Find Out? 5 min

Say: *"We discovered that babies inherit traits from their parents. They look similar but not exactly the same — and that is what makes every animal special!"*

Watch For

Review discoveries. Show one more parent-baby pair and have students identify inherited traits.

Slide 7: STEM Challenge 10 min

Say: *"A zoo needs your help! Look at these two parent animals and draw what their baby might look like. Use the traits you see!"*

Watch For

Give students parent animal cards with clear traits. Students draw predicted babies and explain their choices.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Parent Traits
- **INTERNAL:** Baby Traits

Relationships

- **Parent Traits to Baby Traits:** POSITIVE (+) — When a parent has a strong trait like spots or long ears, the baby is very likely to show a similar trait. More parent traits passed along means the baby looks more like the parent.

Key Discoveries

- Baby animals look a lot like their parents because traits are inherited
- Baby animals are not EXACTLY like their parents — there are always small differences
- Traits like fur color, ear shape, and body size are passed from parents to babies

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Baby animals are exact copies of their parents"	Babies are SIMILAR to their parents but never exactly the same. Just like brothers and sisters look alike but are not identical, baby animals get a mix of traits that makes each one unique.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Baby animals choose what they look like"	Animals cannot choose their traits. Traits are passed from parents to babies automatically, before the baby is even born. A kitten cannot decide to have stripes if its parents have spots.